

**Your Name: Lilian Wanjiku**

**Complete the missing fields in the tables below:**

| <b>Network/prefix</b> | <b>Subnet mask</b> | <b>Host bits</b> | <b>Number of hosts</b> |
|-----------------------|--------------------|------------------|------------------------|
| 172.16.1.0/24         | 255.255.255.0      | 8                | 254                    |
| 10.1.100.128/26       | 255.255.255.192    | 6                | 62                     |
| 10.1.96.0/19          | 255.255.224.0      | 13               | 8190                   |
| 192.168.1.0/24        | 255.255.255.0      | 8                | 254                    |
| 172.31.0.0/16         | 255.255.0.0        | 16               | 65534                  |
| 10.255.255.252/30     | 255.255.255.252    | 2                | 2                      |
| 172.28.240.0/20       | 255.255.240.0      | 12               | 4094                   |
| 10.44.108.0/22        | 255.255.252.0      | 10               | 1022                   |
| 192.168.100.24/21     | 255.255.248.0      | 11               | 2046                   |
| 172.23.64.0/18        | 255.255.192.0      | 14               | 16382                  |
| 192.168.5.128/25      | 255.255.255.128    | 7                | 126                    |

| <b>Network ID</b> | <b>Required hosts</b> | <b>Host bits needed</b> | <b>Network ID/prefix</b> |
|-------------------|-----------------------|-------------------------|--------------------------|
| 172.16.1.0        | 254                   | 8                       | 172.16.1.0/24            |
| 10.1.100.128      | 62                    | 6                       | 10.1.100.128/26          |

|                |       |    |                   |
|----------------|-------|----|-------------------|
| 10.1.96.0      | 8190  | 13 | 10.1.96.0/19      |
| 192.168.1.0    | 254   | 8  | 192.168.1.0/24    |
| 172.31.0.0     | 65534 | 16 | 172.31.0.0/16     |
| 10.255.255.252 | 2     | 2  | 10.255.255.252/30 |
| 172.28.240.0   | 4094  | 12 | 172.28.240.0/20   |
| 10.44.108.0    | 1022  | 10 | 10.44.108.0/22    |
| 192.168.240.0  | 4094  | 12 | 192.168.240.0/20  |
| 172.23.64.0    | 16382 | 14 | 172.23.64.0/18    |
| 192.168.5.128  | 126   | 7  | 192.168.5.128/25  |